PONMATHISH T

DAY - 08

TASK 1 — NETWORK IDENTITY

Q1. What is the IPv4 Address of your computer?

Answer: 10.150.34.91

Q2. What is the Subnet Mask?

Answer: 255.255.255.0

Q3. What is the Default Gateway?

Answer: 10.150.34.120

Q4. What is the DNS Server?

Answer: 10.150.34.120

Q5. What is the MAC/Physical Address?

Answer:

Q6. Is DHCP Enabled on your computer?

Answer: Yes



TASK 4 — DNS INVESTIGATION

Q1. Which DNS server responded?

Answer: The DNS server that responded is 10.150.94.125. The server name is shown as Unknown.

Q2. Which IP address was returned for google.com?

Answer: The DNS lookup returned the following IP addresses:

- **IPv6:** 2404:6800:4007:830::200e

- **IPv4:** 172.217.24.174

Q3. What is the purpose of DNS?

Answer: DNS (Domain Name System) translates human-readable domain names such as google.com into IP addresses. This allows computers to locate and communicate with the correct server on a network or the Internet.

```
C:\Users\acer>nslookup google.com
Server:  UnKnown
Address:  10.150.94.125

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:830::200e
          172.217.24.174


C:\Users\acer>
C:\Users\acer>tracert google.com

Tracing route to google.com [2404:6800:4007:830::200e]
over a maximum of 30 hops:

  1     3 ms     4 ms     3 ms  2409:40f4:1017:8bad::e3
  2     *        *        *     Request timed out.
  3    43 ms    36 ms    39 ms  2405:200:5218:20:3925::ff06
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    42 ms    33 ms    23 ms  2405:200:801:2300::65c
  7     *        *        *     Request timed out.
  8    63 ms    52 ms    45 ms  2001:4860:1:1::16a
  9    49 ms    47 ms    56 ms  2001:4860:1:1::16a
 10    49 ms    44 ms    38 ms  pnmaaa-au-in-x0e.1e100.net [2404:6800:4007:830::200e]

Trace complete.

C:\Users\acer>
```

TASK 5 — NETWORK PATH INVESTIGATION

1. How many visible hops appeared?
There were **10 visible hops** in the traceroute output.

2. What was the first hop?
The first hop was:
2409:40f4:1017:8bad::e3

3. What device commonly represents the first hop?
The first hop commonly represents the **local router/default gateway**.

4. What is traceroute used for?
Traceroute is used to **identify the path taken by network packets from our computer**

to a destination. It shows the different hops between the source and destination and helps identify **network delays, routing problems, or unreachable hops**.